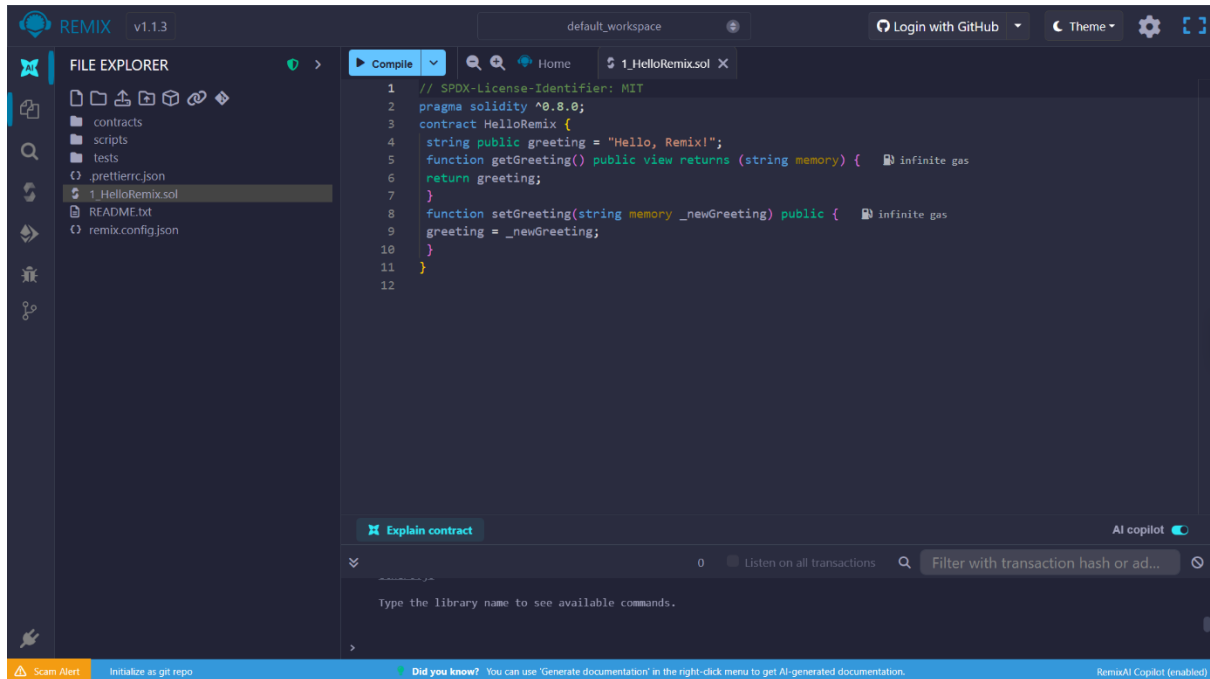


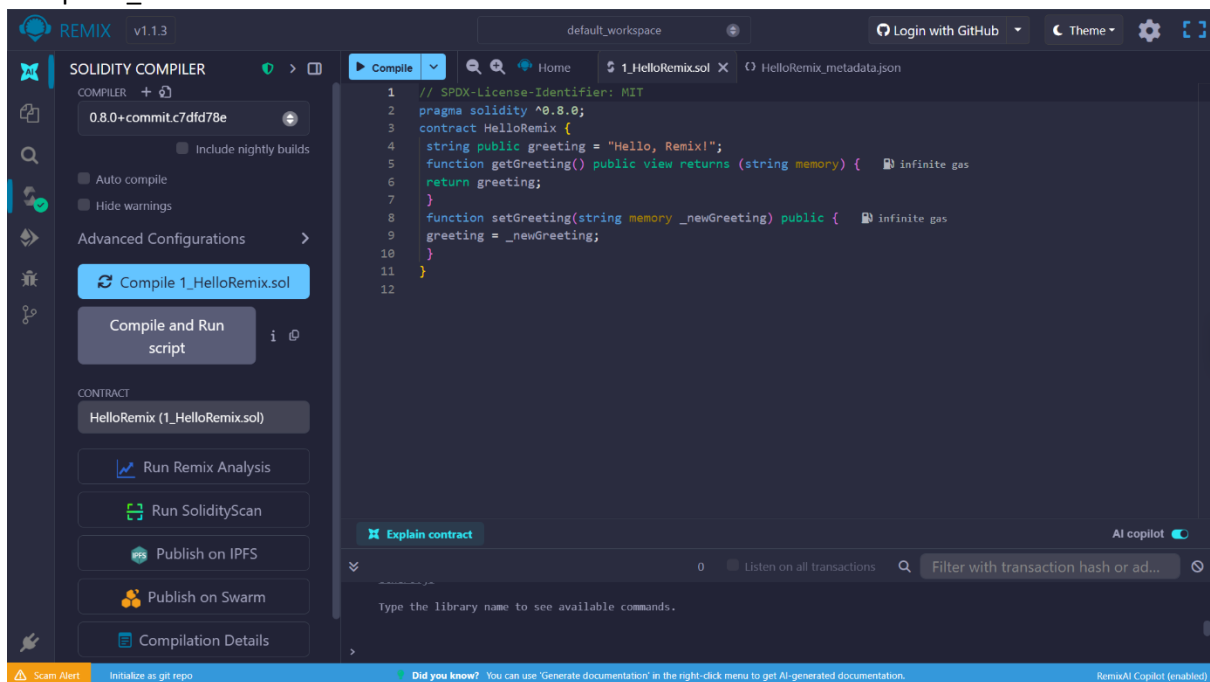
ASSIGNMENT 2

1.

Create 1_HelloRemix.sol



Compile 1_HelloRemix.sol



Deploying the contract

The image displays two screenshots of the Remix IDE interface, showing the process of deploying a Solidity contract.

Top Screenshot: The interface shows the "DEPLOY & RUN TRANSACTIONS" panel on the left. The environment is set to "Remix VM (Prague)". The account is "0x5B3...eddC4 (100 ETH)". The gas limit is set to "Estimated Gas". The contract is "HelloRemix - 1_HelloRemix.sol". The "Deploy" button is highlighted. The main editor shows the Solidity code for the "HelloRemix" contract.

```
1 // SPDX-License-Identifier: MIT
2 pragma solidity ^0.8.0;
3 contract HelloRemix {
4     string public greeting = "Hello, Remix!";
5     function getGreeting() public view returns (string memory) { infinite gas
6         return greeting;
7     }
8     function setGreeting(string memory _newGreeting) public { infinite gas
9         greeting = _newGreeting;
10    }
11 }
12
```

Bottom Screenshot: The interface shows the "Deploy - transact (not payable)" button. The account is "0x5B3...eddC4 (99.9999999999)". The gas limit is set to "Estimated Gas". The contract is "HelloRemix - 1_HelloRemix.sol". The "Deploy" button is highlighted. The main editor shows the same Solidity code. The transaction log at the bottom shows a successful deployment:

```
[vm] from: 0x5B3...eddC4 to: HelloRemix.constructor value: 0 wei data: 0x608...0033 logs: 0
hash: 0xd43...ac13a
```

Click greeting button

The screenshot shows the Remix IDE interface. On the left, the 'DEPLOY & RUN TRANSACTIONS' panel is visible. Under 'Deployed Contracts', the contract 'HELLOREMIX' is listed at address 0xD91...39138. The 'setGreeting' button is highlighted, and the 'greeting' field displays 'Hello, Remix!'. The main editor shows the Solidity code for the contract:

```
1 // SPDX-License-Identifier: MIT
2 pragma solidity ^0.8.0;
3 contract HelloRemix {
4     string public greeting = "Hello, Remix!";
5     function getGreeting() public view returns (string memory) {
6         return greeting;
7     }
8     function setGreeting(string memory _newGreeting) public {
9         greeting = _newGreeting;
10    }
11 }
12
```

The bottom panel shows the transaction details for the 'setGreeting' function call, including the transaction hash and the data field.

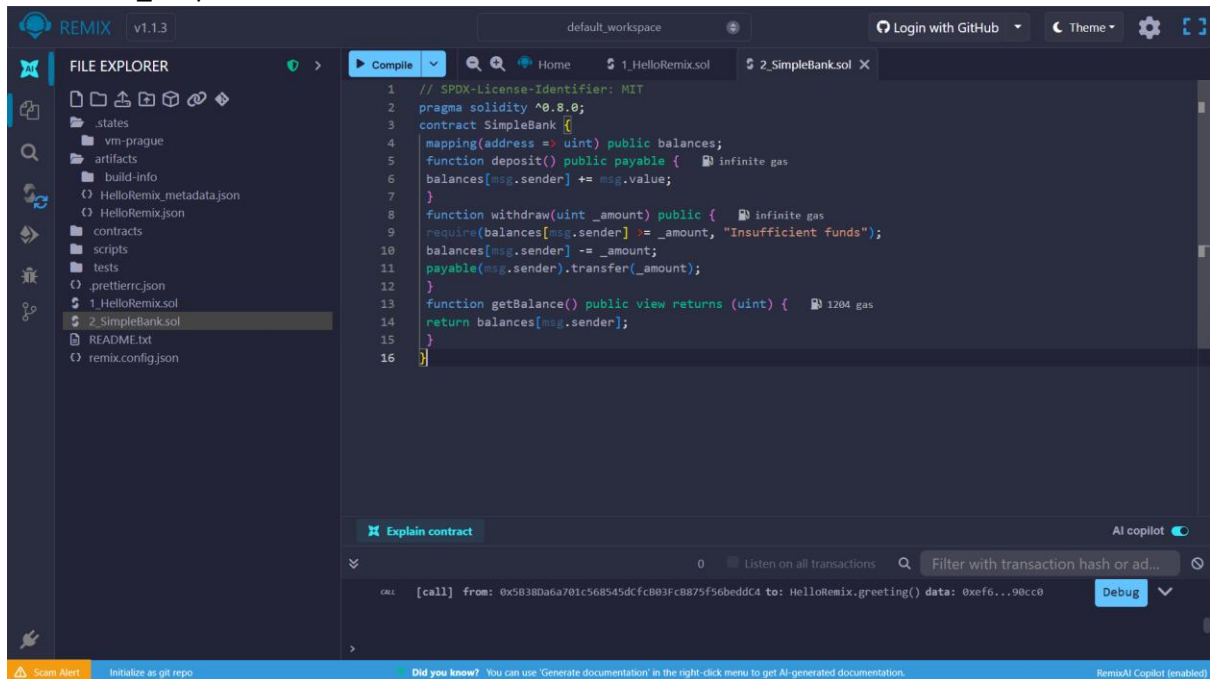
New value

The screenshot shows the Remix IDE interface after the 'setGreeting' function has been called. The 'greeting' field in the 'Deployed Contracts' panel now displays 'Hello, Blockchain Class!'. The main editor shows the same Solidity code as before:

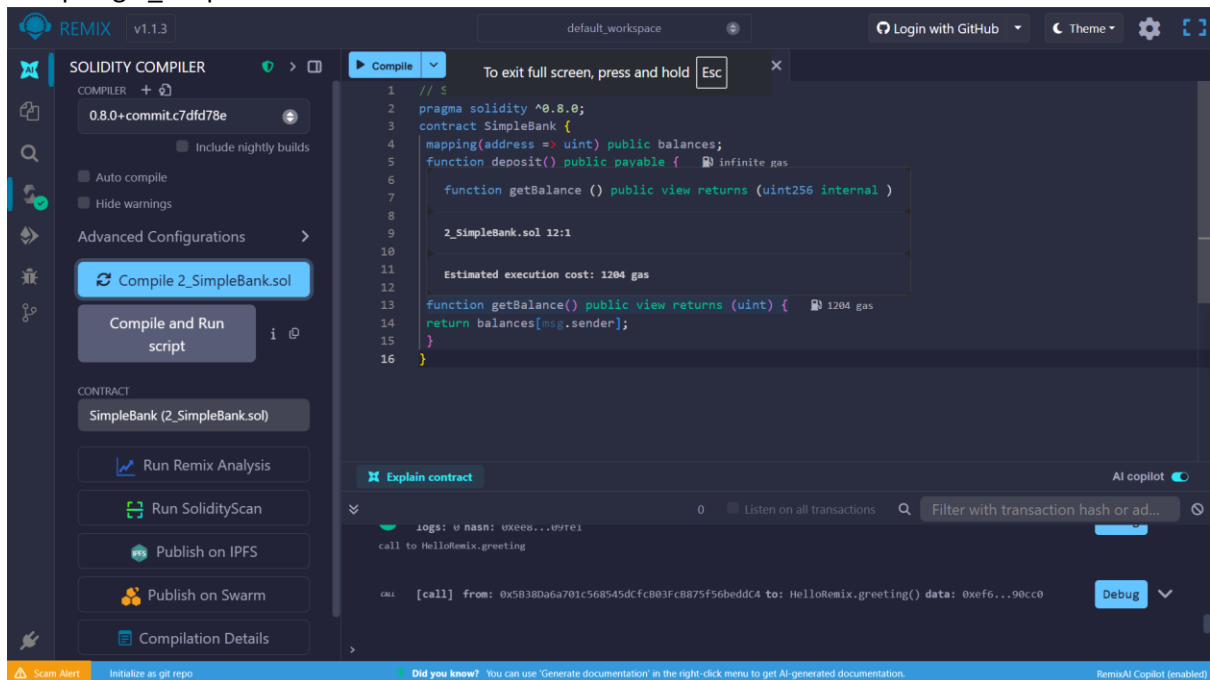
```
1 // SPDX-License-Identifier: MIT
2 pragma solidity ^0.8.0;
3 contract HelloRemix {
4     string public greeting = "Hello, Remix!";
5     function getGreeting() public view returns (string memory) {
6         return greeting;
7     }
8     function setGreeting(string memory _newGreeting) public {
9         greeting = _newGreeting;
10    }
11 }
12
```

The bottom panel shows the transaction details for the 'setGreeting' function call, including the transaction hash and the data field.

2. Create 2_SimpleBank.sol



Compiling 2_SimpleBank.sol



Deploying 2_SimpleBank.sol

The screenshot shows the Remix IDE interface. On the left, the 'DEPLOY & RUN TRANSACTIONS' panel is active, showing a value of 3000000 and a contract named 'SimpleBank - 2_SimpleBank.sol'. The main editor displays the Solidity code for the SimpleBank contract. The bottom panel shows the transaction log with a successful deployment message: '[vm] from: 0x5B3...eddC4 to: SimpleBank.(constructor) value: 0 wei data: 0x608...0033 logs: 0 hash: 0x026...72f24'.

```
1 // SPDX-License-Identifier: MIT
2 pragma solidity ^0.8.0;
3 contract SimpleBank {
4     mapping(address => uint) public balances;
5     function deposit() public payable { infinite gas
6         balances[msg.sender] += msg.value;
7     }
8     function withdraw(uint _amount) public { infinite gas
9         require(balances[msg.sender] >= _amount, "Insufficient funds");
10        balances[msg.sender] -= _amount;
11        payable(msg.sender).transfer(_amount);
12    }
13    function getBalance() public view returns (uint) { 1204 gas
14        return balances[msg.sender];
15    }
16 }
```

Depositing

The screenshot shows the Remix IDE interface. On the left, the 'DEPOSIT' button is active, and the 'getBalance' button is also visible. The main editor displays the Solidity code for the SimpleBank contract. The bottom panel shows the transaction log with a successful deposit message: '[vm] from: 0x5B3...eddC4 to: SimpleBank.deposit() 0xf8e...9f8e8 value: 10000000000000000000 wei data: 0x08e...38db0 logs: 0 hash: 0x2dd...cfcb7 call to SimpleBank.getBalance'.

```
1 // SPDX-License-Identifier: MIT
2 pragma solidity ^0.8.0;
3 contract SimpleBank {
4     mapping(address => uint) public balances;
5     function deposit() public payable { infinite gas
6         balances[msg.sender] += msg.value;
7     }
8     function withdraw(uint _amount) public { infinite gas
9         require(balances[msg.sender] >= _amount, "Insufficient funds");
10        balances[msg.sender] -= _amount;
11        payable(msg.sender).transfer(_amount);
12    }
13    function getBalance() public view returns (uint) { 1204 gas
14        return balances[msg.sender];
15    }
16 }
```


REMIX v1.1.3

default_workspace

Login with GitHub Theme

DEPLOY & RUN TRANSACTIONS

ENVIRONMENT

Remix VM (Prague)

ACCOUNT

0x583...eddC4 (99.49999999999999 ETH)

...eddC4 (99.49999999999999987975 ETH)

...35cb2 (100 ETH)

...C02db (100 ETH)

...caba8 (100 ETH)

...5E7f2 (100 ETH)

...8c372 (100 ETH)

...21678 (100 ETH)

...D1Ff7 (100 ETH)

...E454C (100 ETH)

...C70DC (100 ETH)

...a733c (100 ETH)

...C160C (100 ETH)

...4D2dB (100 ETH)

...40225 (100 ETH)

...92148 (100 ETH)

Compile

1_HelloRemix.sol 2_SimpleBank.sol

```

1 // SPDX-License-Identifier: MIT
2 pragma solidity ^0.8.0;
3 contract SimpleBank {
4     mapping(address => uint) public balances;
5     function deposit() public payable {
6         balances[msg.sender] += msg.value;
7     }
8     function withdraw(uint _amount) public {
9         require(balances[msg.sender] >= _amount, "Insufficient funds");
10        balances[msg.sender] -= _amount;
11        payable(msg.sender).transfer(_amount);
12    }
13    function getBalance() public view returns (uint) {
14        return balances[msg.sender];
15    }
16 }

```

Explain contract

AI copilot

0 Listen on all transactions

Filter with transaction hash or ad...

transact to SimpleBank.withdraw pending ...

[vm] from: 0x583...eddC4 to: SimpleBank.withdraw(uint256) 0xf8e...9f8e8 value: 0 wei data: 0x2e1...20000

logs: 0 hash: 0x51f...4fc7f

Debug

Scam Alert

Initialize as git repo

Did you know? You can use 'Generate documentation' in the right-click menu to get AI-generated documentation.

RemixAI Copilot (enabled)

Withdrawal

REMIX v1.1.3

default_workspace

Login with GitHub Theme

DEPLOY & RUN TRANSACTIONS

At Address

Load contract from Address

Transactions recorded

1

Deployed Contracts

1

SIMPLEBANK AT 0XF8E...9F8E8

Balance: 0.5 ETH

deposit

withdraw 5000000000000000000

balances address

getBalance

0. uint256: 5000000000000000000

Low level interactions

1

CALLDATA

Transact

Compile

1_HelloRemix.sol 2_SimpleBank.sol

```

1 // SPDX-License-Identifier: MIT
2 pragma solidity ^0.8.0;
3 contract SimpleBank {
4     mapping(address => uint) public balances;
5     function deposit() public payable {
6         balances[msg.sender] += msg.value;
7     }
8     function withdraw(uint _amount) public {
9         require(balances[msg.sender] >= _amount, "Insufficient funds");
10        balances[msg.sender] -= _amount;
11        payable(msg.sender).transfer(_amount);
12    }
13    function getBalance() public view returns (uint) {
14        return balances[msg.sender];
15    }
16 }

```

Explain contract

AI copilot

0 Listen on all transactions

Filter with transaction hash or ad...

Aug: 0 msg: 0x2e1...41c7f

call to SimpleBank.getBalance

call [call] from: 0x58380a6a701c568545dcfc803fc8875f56beddC4 to: SimpleBank.getBalance() data: 0x120...65fe0

Debug

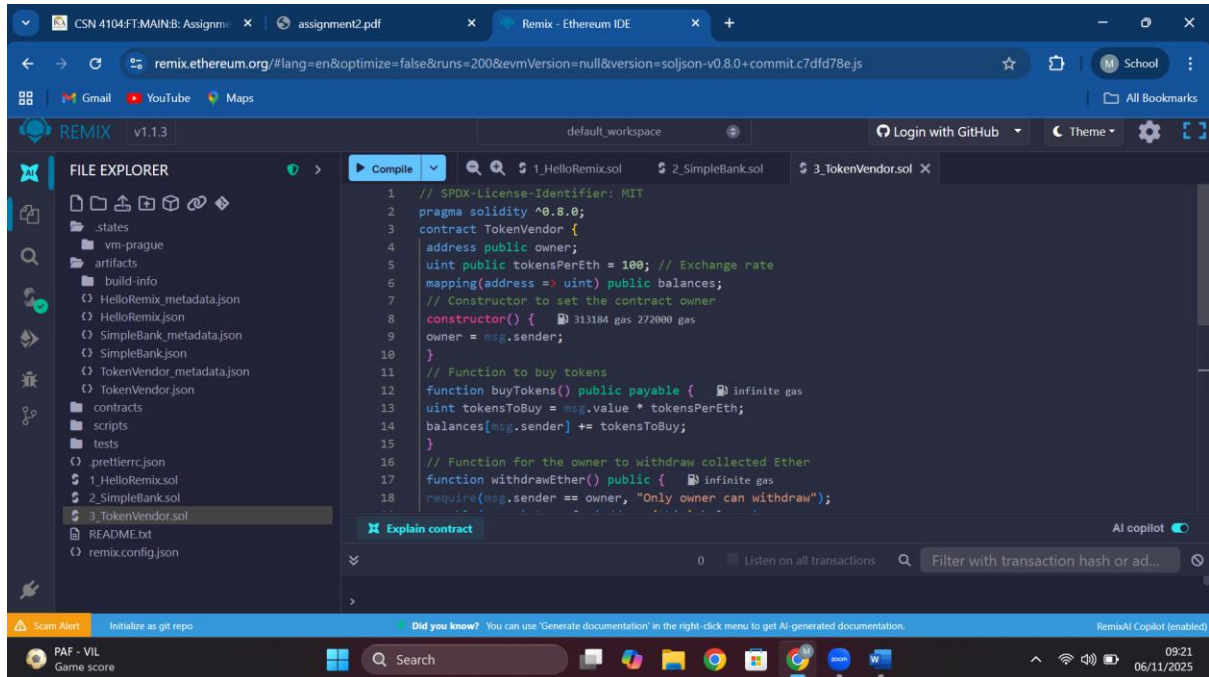
Scam Alert

Initialize as git repo

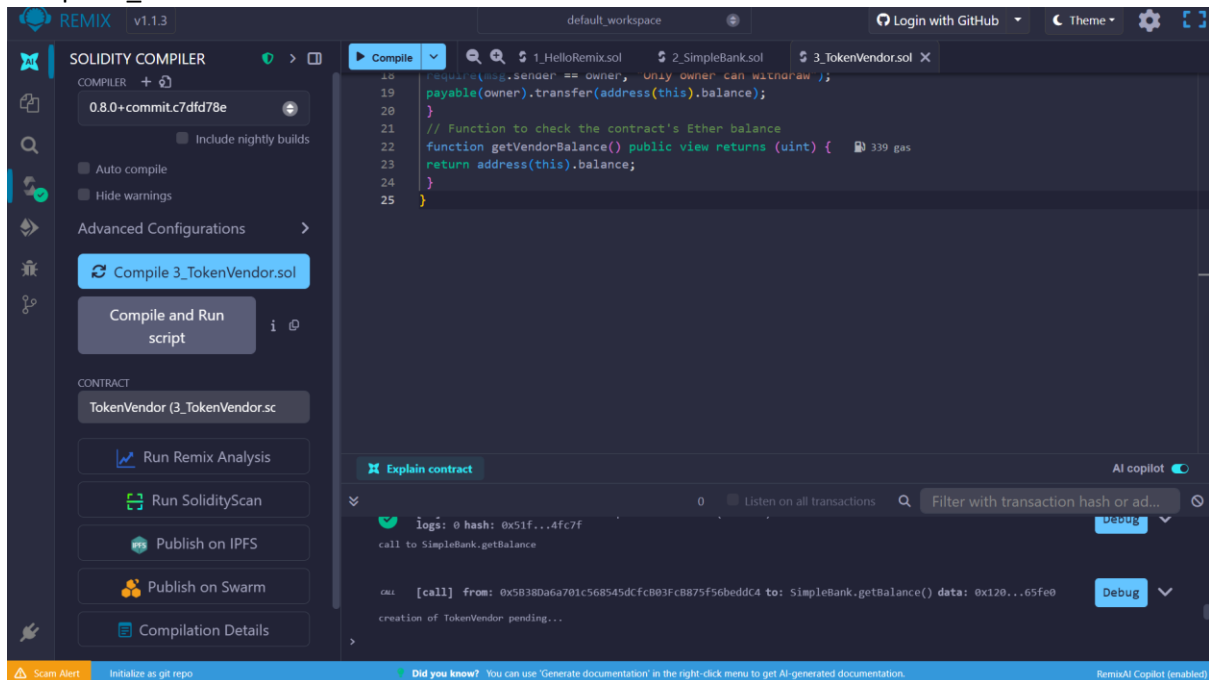
Did you know? You can use 'Generate documentation' in the right-click menu to get AI-generated documentation.

RemixAI Copilot (enabled)

3. Create 3_TokenVendor.sol



Compile 3_TokenVendor.sol



DEPLOY & RUN TRANSACTIONS

ENVIRONMENT: Remix VM (Prague)

ACCOUNT: 0x5B3...eddC4 (99.4999999999)

GAS LIMIT: Estimated Gas (3000000)

VALUE: 0 Ether

CONTRACT: TokenVendor - 3_TokenVendor.sol

Deploy

At Address

Transactions recorded: 1

```

18 require(msg.sender == owner, "Only owner can withdraw");
19 payable(owner).transfer(address(this).balance);
20 }
21 // Function to check the contract's Ether balance
22 function getVendorBalance() public view returns (uint) {
23     return address(this).balance;
24 }
25 }

```

creation of TokenVendor pending...

[vm] from: 0x5B3...eddC4 to: TokenVendor.(constructor) value: 0 wei data: 0x608...00033 logs: 0 hash: 0xf81...dd5bf

Deposit 0.1 ether

DEPLOYED CONTRACTS

TRANSACTIONS

Transactions recorded: 13

Deployed Contracts: 1

TOKENVENDOR AT 0xD91...391

Balance: 0.1 ETH

buyTokens

withdrawEther

balances

getVendorBala...

owner

tokensPerEth

Low level interactions

CALLDATA

Transact

```

1 // SPDX-License-Identifier: MIT
2 pragma solidity ^0.8.0;
3 contract TokenVendor {
4     address public owner;
5     uint public tokensPerEth = 100; // Exchange rate
6     mapping(address => uint) public balances;
7     // Constructor to set the contract owner
8     constructor() {
9         owner = msg.sender;
10    }
11    // Function to buy tokens
12    function buyTokens() public payable {
13        uint tokensToBuy = msg.value * tokensPerEth;
14        balances[msg.sender] += tokensToBuy;
15    }
16    // Function for the owner to withdraw collected Ether
17    function withdrawEther() public {
18        require(msg.sender == owner, "Only owner can withdraw");
19        payable(owner).transfer(address(this).balance);
20    }
21 }

```

[vm] from: 0x5B3...eddC4 to: TokenVendor.(constructor) value: 0 wei data: 0x608...00033 logs: 0 hash: 0xd29...48256

transact to TokenVendor.buyTokens pending ...

[vm] from: 0x5B3...eddC4 to: TokenVendor.buyTokens() 0xd91...39138 value: 1000000000000000000 wei data: 0xd0f...ebe4c logs: 0 hash: 0x7c7...4f963

Token balance

The screenshot shows the REMIX IDE v1.1.3 interface. On the left, the 'DEPLOY & RUN TRANSACTIONS' sidebar is active. It displays the 'TokenVendor - 3_TokenVendor.sol' contract with the 'istanbul' EVM version selected. The 'Deploy' button is visible. Below, the 'Transactions recorded' section shows 15 transactions. The 'Deployed Contracts' section lists the 'TOKENVENDOR AT 0XD91...391' contract with a balance of 0.1 ETH. It includes buttons for 'buyTokens', 'withdrawEther', and 'balances'. The 'balances' button is selected, showing a value of '0xd9145CCES2D386254917' and a unit of 'uint256: 10'. The main editor shows the Solidity code for the 'TokenVendor' contract, which includes a constructor, a 'buyTokens' function, and a 'withdrawEther' function. The bottom status bar indicates 'Scan Alert' and 'Initialize as git repo'.

Test access control

This screenshot shows the REMIX IDE v1.1.3 interface after the execution of the 'withdrawEther' function. The 'DEPLOYED CONTRACTS' sidebar on the left shows the 'TOKENVENDOR AT 0XD91...391' contract with a balance of 0 ETH. The 'balances' button is selected, showing the same address and unit. The 'getVendorBala...' button is also visible. The main editor shows the Solidity code for the 'TokenVendor' contract. The bottom status bar indicates 'Scan Alert' and 'Initialize as git repo'. The 'Explain contract' panel on the right shows the transaction details for the 'withdrawEther' function call, including the transaction hash, block hash, block number, from address, to address, gas, transaction cost, execution cost, input, and output.

4. Re-opening and modifying 2_SimpleBank.sol

The screenshot shows the Remix IDE interface. On the left, the 'SOLIDITY COMPILER' panel is active, displaying the compiler version '0.8.0+commit.c7dfd78e' and various options like 'Include nightly builds', 'Auto compile', and 'Hide warnings'. Below these are buttons for 'Compile 2_SimpleBank.sol', 'Compile and Run script', 'Run Remix Analysis', 'Run SolidityScan', 'Publish on IPFS', 'Publish on Swarm', and 'Compilation Details'. The main editor displays the Solidity code for '2_SimpleBank.sol'. The code defines a contract with a mapping for balances, events for DepositMade and WithdrawalMade, and functions for deposit, withdraw, and getBalance. The deposit function is highlighted. At the bottom, the 'Explain contract' panel shows a transaction call: '[call] from: 0xA8B483F64d9C6d1EcF9B849Ae677D03315835cb2 to: TokenVendor.owner() data: 0x8da...5cb5b'. The bottom status bar indicates 'RemixAI Copilot (enabled)'.

Deposit of 0.5 ether

The screenshot shows the Remix IDE interface after deploying the contract. On the left, the 'DEPLOY & RUN TRANSACTIONS' panel is active, showing the contract 'SimpleBank - 2_SimpleBank.sol' with 'evm version: istanbul'. Below are buttons for 'Deploy', 'At Address', and 'Load contract from Address'. The 'Transactions recorded' section shows a list of transactions. The 'Deployed Contracts' section shows the contract 'SIMPLEBANK AT 0XD2A...FD006' with a balance of '0.5 ETH'. Below this are buttons for 'deposit', 'withdraw', 'balances', and 'getBalance'. The 'Low level interactions' section shows a 'CALLDATA' field and a 'Transact' button. The main editor displays the Solidity code for '2_SimpleBank.sol'. The code defines a contract with a mapping for balances, events for DepositMade and WithdrawalMade, and functions for deposit, withdraw, and getBalance. The deposit function is highlighted. At the bottom, the 'Explain contract' panel shows a transaction call: '[vm] from: 0x5B3...eddC4 to: SimpleBank.deposit() 0xd2a...fd005 value: 10000000000000000 wei data: 0xd0e...30db0 logs: 1 hash: 0xd82...3ef4f'. The bottom status bar indicates 'RemixAI Copilot (enabled)'.

Withdrawal of 0.2 ether

[illegible]

5. Creating 5_SimpleAuction.sol

The screenshot shows the Remix IDE interface. On the left, the 'FILE EXPLORER' panel lists files in the 'contracts' directory, with '5_SimpleAuction.sol' highlighted. The main editor window shows the Solidity code for the 'SimpleAuction' contract. The code includes a SPDX license header, pragma statement, and a contract definition with various features and events. The bottom status bar indicates 'AI copilot' is active and 'RemixAI Copilot (enabled)'.

```

1 // SPDX-License-Identifier: MIT
2 pragma solidity ^0.8.0;
3
4 /*
5  SimpleAuction contract
6  -----
7  Features:
8  - Owner creates the auction.
9  - Users can place bids by sending Ether.
10 - Automatically refunds the previous highest bidder when outbid.
11 - Owner can end the auction and withdraw the highest bid.
12 - Emits events for new bids and when the auction ends.
13 */
14
15 contract SimpleAuction {
16     address public owner;
17     address public highestBidder;
18     uint public highestBid;
19     bool public auctionEnded;
20
21     // Track pending refunds for previous bidders
22     mapping(address => uint) public pendingReturns;
23
24     // Events for off-chain logging
25     event NewHighestBid(address indexed bidder, uint amount);
26     event AuctionEnded(address winner, uint amount);

```

REMIX v1.1.3 default_workspace Login with GitHub Theme

FILE EXPLORER

- .states
- artifacts
- contracts
- scripts
- tests
- .prettierrc.json
- 1_HelloRemix.sol
- 2_SimpleBank.sol
- 3_TokenVendor.sol
- 5_SimpleAuction.sol
- README.txt
- remix.config.json

26 event AuctionEnded(address winner, uint amount);

27

28

29 // Constructor to set the auction owner

30 constructor() { 556234 gas 534800 gas

31 | owner = msg.sender;

32 }

33

34 // Function to place a bid

35 function bid() public payable { infinite gas

36 require(!auctionEnded, "Auction already ended");

37 require(msg.value > highestBid, "There already is a higher bid");

38

39 // Refund the previous highest bidder

40 if (highestBid != 0) {

41 | pendingReturns[highestBidder] += highestBid;

42 }

43

44 // Update the highest bid

45 highestBidder = msg.sender;

46 highestBid = msg.value;

47

48 emit NewHighestBid(msg.sender, msg.value);

49 }

50

51 // Function to withdraw refunds (in case bidder was outbid)

52 function withdraw() public { infinite gas

53

54

55

56

57

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Explain contract AI copilot

0 Listen on all transactions Filter with transaction hash or ad...

REMIX v1.1.3 default_workspace Login with GitHub Theme

FILE EXPLORER

- .states
- artifacts
- contracts
- scripts
- tests
- .prettierrc.json
- 1_HelloRemix.sol
- 2_SimpleBank.sol
- 3_TokenVendor.sol
- 5_SimpleAuction.sol
- README.txt
- remix.config.json

50

51 // Function to withdraw refunds (in case bidder was outbid)

52 function withdraw() public { infinite gas

53 uint amount = pendingReturns[msg.sender];

54 require(amount > 0, "No funds to withdraw");

55

56 // Reset before sending to prevent reentrancy

57 pendingReturns[msg.sender] = 0;

58

59 payable(msg.sender).transfer(amount);

60 }

61

62 // Function to end the auction (only owner)

63 function endAuction() public { infinite gas

64 require(msg.sender == owner, "Only owner can end the auction");

65 require(!auctionEnded, "Auction already ended");

66

67 auctionEnded = true;

68

69 // Send highest bid to owner

70 payable(owner).transfer(highestBid);

71

72 emit AuctionEnded(highestBidder, highestBid);

73 }

74

75 // Helper function to check contract balance

76 function getContractBalance() public view returns (uint) { 406 gas

77

78

79

80

81

82

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85

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97

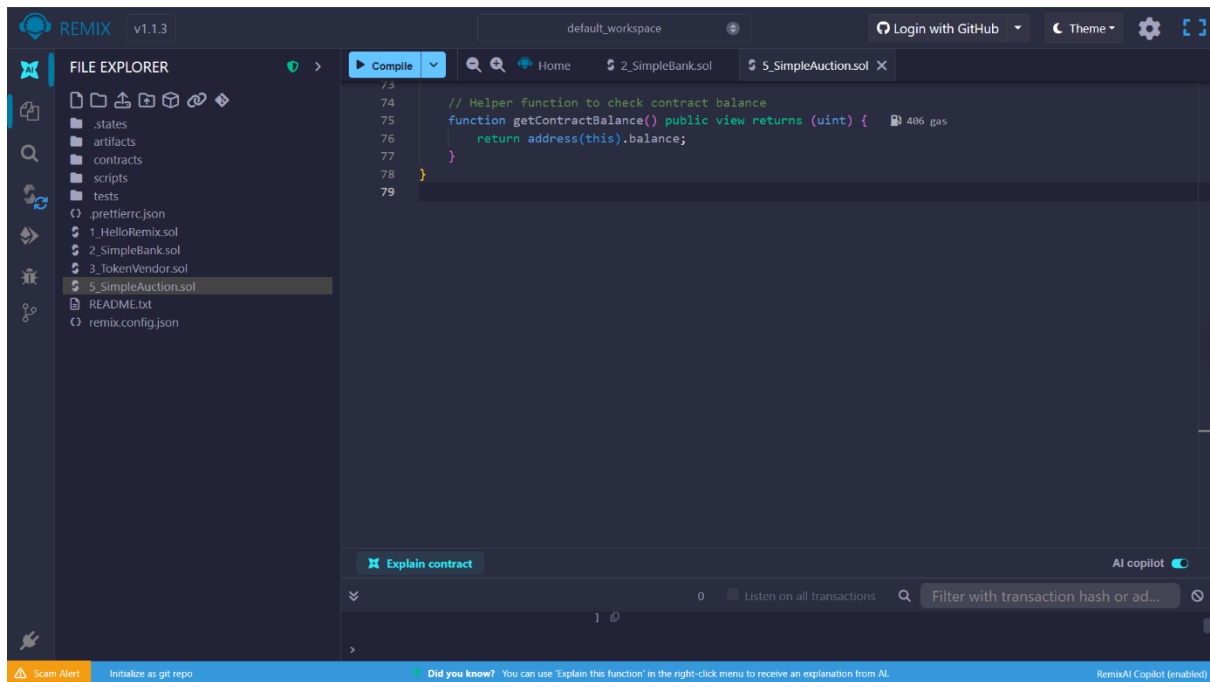
98

99

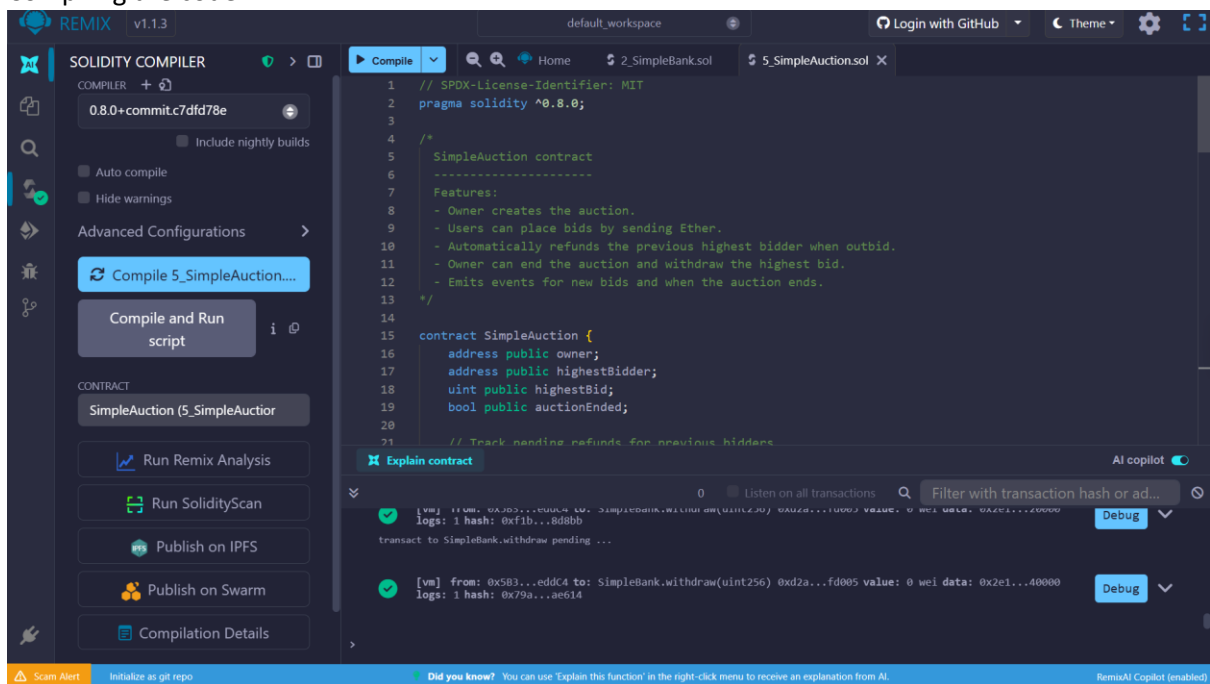
100

Explain contract AI copilot

0 Listen on all transactions Filter with transaction hash or ad...



Compiling the code



REMIX v1.1.3 default_workspace Login with GitHub Theme

DEPLOY & RUN TRANSACTIONS

SIMPLEAUCTION AT 0XC66...99

Balance: 0 ETH

bid

endAuction

withdraw

auctionEnded

getContractBal...

highestBid

highestBidder

owner

pendingReturns address

Low level interactions

CALLDATA

Transact

Compile

2_SimpleBank.sol 5_SimpleAuction.sol

```

1 // SPDX-License-Identifier: MIT
2 pragma solidity ^0.8.0;
3
4 /*
5  SimpleAuction contract
6  -----
7  Features:
8  - Owner creates the auction.
9  - Users can place bids by sending Ether.
10 - Automatically refunds the previous highest bidder when outbid.
11 - Owner can end the auction and withdraw the highest bid.
12 - Emits events for new bids and when the auction ends.
13 */
14
15 contract SimpleAuction {
16     address public owner;
17     address public highestBidder;
18     uint public highestBid;
19     bool public auctionEnded;
20
21     // Track pending refunds for previous bidders

```

Explain contract

AI copilot

Listen on all transactions

Filter with transaction hash or address

[vm] from: 0x608...00033 to: SimpleAuction.(constructor) value: 0 wei data: 0x608...00033 logs: 0 creation of SimpleAuction pending...

[vm] from: 0x5B3...eddC4 to: SimpleAuction.(constructor) value: 0 wei data: 0x608...00033 logs: 0 hash: 0xa25...8577b

Scam Alert Initialize as git repo Did you know? You can use 'Explain this function' in the right-click menu to receive an explanation from AI. RemixAI Copilot (enabled)

Bid 1 of 0.5 ether

REMIX v1.1.3 default_workspace Login with GitHub Theme

DEPLOY & RUN TRANSACTIONS

At Address Load contract from Address

Transactions recorded 19

Deployed Contracts 2

TOKENVENDOR AT 0XD91...391

SIMPLEAUCTION AT 0X652...BA

Balance: 0.5 ETH

bid

endAuction

withdraw

auctionEnded

getContractBal...

highestBid

highestBidder

owner

pendingReturns address

Compile

2_SimpleBank.sol 5_SimpleAuction.sol 3_TokenVendor.sol

```

1 // SPDX-License-Identifier: MIT
2 pragma solidity ^0.8.0;
3
4 /*
5  SimpleAuction contract

```

Listen on all transactions

Filter with transaction hash or address

[vm] from: 0x199...8aeef to: SimpleAuction.bid() 0x652...bA595 value: 500000000000000000 wei data: 0x199...8aeef logs: 1 hash: 0xe04...fbf25

status 0x1 Transaction mined and execution succeed

transaction hash 0xe0448e9576a684dc8d19bf0992c57be59f683c0b0d6f20b05024135a7efb725

block hash 0x5658752195bb8c4779c27cb6b1e6a99f1b1fdeea34b696e27174707a71ef87e

block number 8

from 0x4b483f64d9c6d1fc9b849a677d0315835cb2

to SimpleAuction.bid() 0x652c98cc53e705e1d96e2455e6184a879bA595

gas 79739 gas

transaction cost 69338 gas

execution cost 48274 gas

input 0x199...8aeef

output 0x

Scam Alert Initialize as git repo Did you know? You can use 'Explain this function' in the right-click menu to receive an explanation from AI. RemixAI Copilot (enabled)

The screenshot shows the Remix IDE interface. On the left, the 'DEPLOY & RUN TRANSACTIONS' panel is active, displaying the 'SIMPLEAUCTION' contract at address 0x652...BA. The contract's balance is 0.5 ETH. A list of buttons for interacting with the contract is visible, including 'bid', 'endAuction', 'withdraw', 'auctionEnded', 'getContractBal...', 'highestBid', 'highestBidder', 'owner', and 'pendingReturns'. The 'bid' button is highlighted in red. The main editor shows the Solidity code for the 'SimpleAuction' contract, which is a simple auction mechanism. The transaction logs on the right show a successful transaction with a value of 5000000000000000000 wei.

Highest bid for bid 1

This screenshot shows the Remix IDE interface after a transaction. The 'DEPLOY & RUN TRANSACTIONS' panel on the left shows the 'SIMPLEAUCTION' contract at address 0x652...BA. The 'bid' button is highlighted in red. The main editor shows the Solidity code for the 'SimpleAuction' contract. The transaction logs on the right show a successful transaction with a value of 5000000000000000000 wei. The logs also show the transaction hash and the transaction index.

Highest bidder for bid 1

The screenshot shows the Remix IDE interface. On the left, the 'DEPLOY & RUN TRANSACTIONS' panel displays the 'SimpleAuction' contract with a balance of 0.5 ETH. The 'highestBidder' function is highlighted. The main editor shows the Solidity code for the 'SimpleAuction' contract. The right panel shows the transaction details for the 'highestBidder' function call, including the transaction hash, block hash, and execution cost.

```
1 // SPDX-License-Identifier: MIT
2 pragma solidity ^0.8.0;
3
4 /*
5  SimpleAuction contract
```

Transaction details:

- from: 0xAb8483F64d9C6d1EcF9b849Ae677dD3315835cb2
- to: SimpleAuction.highestBidder() 0x652c9AcC53e765e1d96e2455E618dAaB79bA595
- execution cost: 2558 gas (Cost only applies when called by a contract)
- input: 0x91f...90157
- output: 0x00
- decoded input: {}
- decoded output: {"0": "address: 0xAb8483F64d9C6d1EcF9b849Ae677dD3315835cb2"}
- logs: []
- raw logs: []

Bid 2 of 0.8 ether

The screenshot shows the Remix IDE interface. On the left, the 'DEPLOY & RUN TRANSACTIONS' panel displays the 'SimpleAuction' contract with a balance of 1.3 ETH. The 'highestBidder' function is highlighted. The main editor shows the Solidity code for the 'SimpleAuction' contract. The right panel shows the transaction details for the 'highestBidder' function call, including the transaction hash, block hash, and execution cost.

```
1 // SPDX-License-Identifier: MIT
2 pragma solidity ^0.8.0;
3
4 /*
5  SimpleAuction contract
```

Transaction details:

- status: 0x1 Transaction mined and execution succeed
- transaction hash: 0xdce6aa90cc04f23e61842bbe23d6ae08b37b92eda63a4388c12c276f1d0e912
- block hash: 0xa4ba6adcc1b493103c346a879c9d2fea87a21bc73a47e2829f4e977e69db
- block number: 9
- from: 0xAb8483F64d9C6d1EcF9b849Ae677dD3315835cb2
- to: SimpleAuction.bid() 0xd51c9AcC53e765e1d96e2455E618dAaB79bA595
- gas: 63233 gas
- transaction cost: 54385 gas
- execution cost: 33921 gas
- input: 0x199...8aeef
- output: 0x

Highest bid for bid 2

[illegible]

Highest bidder for bid 2

The screenshot displays the Remix IDE interface, which is used for developing smart contracts. The top bar shows the version 'v1.1.3' and options to login with GitHub or change the theme.

The left sidebar contains navigation icons and a section titled 'DEPLOY & RUN TRANSACTIONS'. Under this section, there are tabs for different transactions:

- TOKENVENDOR AT 0xD91...391
- SIMPLEAUCTION AT 0x652...BA

The 'SIMPLEAUCTION AT 0x652...BA' tab is active, showing a balance of 1.3 ETH and several transaction buttons: bid, endAuction, withdraw, auctionEnded, getContractBal..., highestBid, and highestBidder. Below these buttons, there are input fields for 'uint256' and 'address', and a dropdown menu for 'owner'. At the bottom of the sidebar, there are sections for 'Low level interactions' and 'CALLDATA'.

The main workspace area shows the Solidity code for 'SimpleAuction.sol':`// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;

/*
 SimpleAuction contract
*/`

Below the code editor, there is a section for transaction details. It shows a call from '0xAxB483F64d9C6d1EcF9B849Ae677dD3315835cb2' to 'SimpleAuction.highestBidder()'. The transaction data is '0x91f...90157'. The execution cost is 2558 gas. The output is '0x000000000000000000000000ab8483f64d9c6d1ecf9b849ae677dd3315835cb2'. The decoded output is '0': 'address: 0xAxB483F64d9C6d1EcF9B849Ae677dD3315835cb2'.

The bottom status bar includes a 'Scam Alert' icon, a message about initializing as a git repo, a 'Did you know?' tip, and the 'RemixAI Copilot (enabled)' logo.

Refund to the first bidder

REMIXv1.1.3

default_workspace

Login with GitHubTheme

DEPLOY & RUN TRANSACTIONS

Transactions recorded28

Deployed Contracts2

TOKENVENDOR AT 0xD91...391

SIMPLEAUCTION AT 0x652...BA

Balance: 0.8 ETH

bid

endAuction

withdraw

auctionEnded

getContractBal...

0: uint256: 1300000000000000000

highestBid

0: uint256: 800000000000000000

highestBidder

0: address: 0xab84b3f64d9c6d1ecf9b849ae677dd3315835cb2

Compile

2_SimpleBank.sol

5_SimpleAuction.sol

3_TokenVendor.sol

1// SPDX-License-Identifier: MIT

2pragma solidity ^0.8.0;

3

4/*

5SimpleAuction contract

0Listen on all transactions

Filter with transaction hash or ad...

[vm] from: 0xab8...35cb2 to: SimpleAuction.withdraw() 0x652...ba595 value: 0 wei data: 0x3cc...fd60b logs: 0

hash: 0x701...132ab

status0x1 Transaction mined and execution succeed

transaction hash0x701e2161d0912ef07063c95c4de12a3adec6de955e29cce5320e2c70356132ab

block hash0xe6756ba21473be74700284d783dbf75b2fa62e8fbaeef732e9a4d8656d129ed

block number17

from0xab84b3f64d9c6d1ecf9b849ae677dd3315835cb2

toSimpleAuction.withdraw() 0x652c9acc53e765e1d96e2455e618daa879ba595

gas43912 gas

transaction cost28584 gas

execution cost12320 gas

input0x3cc...fd60b

output0x

Scam Alert

Initialize as git repo

Did you know? You can use 'Explain this function' in the right-click menu to receive an explanation from AI.

RemixAI Copilot (enabled)

REMIXv1.1.3

default_workspace

Login with GitHubTheme

DEPLOY & RUN TRANSACTIONS

Transactions recorded28

Deployed Contracts2

TOKENVENDOR AT 0xD91...391

SIMPLEAUCTION AT 0x652...BA

Balance: 0.8 ETH

bid

endAuction

withdraw

auctionEnded

getContractBal...

0: uint256: 1300000000000000000

highestBid

0: uint256: 800000000000000000

highestBidder

0: address: 0xab84b3f64d9c6d1ecf9b849ae677dd3315835cb2

Compile

2_SimpleBank.sol

5_SimpleAuction.sol

3_TokenVendor.sol

1// SPDX-License-Identifier: MIT

2pragma solidity ^0.8.0;

3

4/*

5SimpleAuction contract

0Listen on all transactions

Filter with transaction hash or ad...

from0xab84b3f64d9c6d1ecf9b849ae677dd3315835cb2

toSimpleAuction.withdraw() 0x652c9acc53e765e1d96e2455e618daa879ba595

gas43912 gas

transaction cost28584 gas

execution cost12320 gas

input0x3cc...fd60b

output0x

decoded input()

decoded output()

logs[]

raw logs[]

Scam Alert

Initialize as git repo

Did you know? You can use 'Explain this function' in the right-click menu to receive an explanation from AI.

RemixAI Copilot (enabled)

End of auction

REMX v1.1.3 default_workspace Login with GitHub Theme

DEPLOY & RUN TRANSACTIONS

Transactions recorded 29

Deployed Contracts 2

TOKENVENDOR AT 0XD91...391

SIMPLEAUCTION AT 0X652...BA

Balance: 0 ETH

bid

endAuction

withdraw

auctionEnded

getContractBal...

uint256: 1300000000000000000

uint256: 8000000000000000000

highestBidder

address: 0xAb8483F64d9C6d1EcF9b849Ae677dD3315835cb2

```
1 // SPDX-License-Identifier: MIT
2 pragma solidity ^0.8.0;
3
4 /*
5  SimpleAuction contract
6  */
```

0 Listen on all transactions Filter with transaction hash or ad...

[vm] from: 0xab8...35cb2 to: SimpleAuction.endAuction() 0x652...ba595 value: 0 wei data: 0xfe6...7a54b logs: 1 hash: 0xdfe...df875

status 0x1 Transaction mined and execution succeed

transaction hash 0xdfec64ec18ce398072e5608e5a30ef3f44a7f84d859af623f63f0becea1df875

block hash 0x1ff8accfd1163460003ac8db342e1251ea3ec9c93b5af36c0bb5c2d21211e48

block number 18

from 0xab8483f64d9c6d1ecf9b849ae677dd3315835cb2

to SimpleAuction.endAuction() 0x652c9acc53e765e1d96e2455e618da879ba595

gas 67572 gas

transaction cost 58758 gas

execution cost 37694 gas

input 0xfe6...7a54b

output 0x

Scam Alert Initialize as git repo Did you know? You can use 'Explain this function' in the right-click menu to receive an explanation from AI. RemixAI Copilot (enabled)

REMX v1.1.3 default_workspace Login with GitHub Theme

DEPLOY & RUN TRANSACTIONS

Transactions recorded 29

Deployed Contracts 2

TOKENVENDOR AT 0XD91...391

SIMPLEAUCTION AT 0X652...BA

Balance: 0 ETH

bid

endAuction

withdraw

auctionEnded

getContractBal...

uint256: 1300000000000000000

uint256: 8000000000000000000

highestBidder

address: 0xAb8483F64d9C6d1EcF9b849Ae677dD3315835cb2

```
1 // SPDX-License-Identifier: MIT
2 pragma solidity ^0.8.0;
3
4 /*
5  SimpleAuction contract
6  */
```

0 Listen on all transactions Filter with transaction hash or ad...

logs

raw logs

Scam Alert Initialize as git repo Did you know? You can use 'Explain this function' in the right-click menu to receive an explanation from AI. RemixAI Copilot (enabled)

AuctionEnded event after ending the auction

The screenshot displays the Remix IDE interface with the following components:

- Top Bar:** Shows the Remix logo, version 1.1.3, and the workspace name "default_workspace". It includes a "Login with GitHub" button and a "Theme" dropdown.
- Left Panel (Deploy & Run Transactions):**
 - Shows the deployment status: "TOKENVENDOR AT 0XD91...391:" and "SIMPLEAUCTION AT 0X652...BA".
 - Displays the account balance: "Balance: 0 ETH".
 - Contains buttons for "bid", "endAuction", "withdraw", "auctionEnded", "getContractBal...", "highestBid", "highestBidder", and "owner".
 - Shows the contract address: "address: 0xAb8483F64d9C6d1EcF9b849Ae677dD3315835cb2".
- Editor:** Displays the Solidity code for the `SimpleAuction` contract:

```
1 // SPDX-License-Identifier: MIT
2 pragma solidity ^0.8.0;
3
4 /*
5  SimpleAuction contract
```
- Right Panel (Transaction Log):**
 - Shows a list of transactions with a search bar "Filter with transaction hash or address".
 - Displays a transaction with the following details:
 - From: `0xab8...35cb2` to: `SimpleAuction.bid()`
 - To: `0x652...ba595`
 - Value: `0 wei`
 - Data: `0x199...8aef`
 - Logs: `0`
 - Shows the transaction status: "transaction to SimpleAuction.bid pending ...".
 - Displays an error message: "[vm] from: 0xab8...35cb2 to: SimpleAuction.bid() 0x652...ba595 value: 0 wei data: 0x199...8aef logs: 0 hash: 0x18a...08477".
 - Shows the transaction status: "transaction to SimpleAuction.bid errored: Error occurred: revert".
 - Displays the reason for the revert: "The transaction has been reverted to the initial state. Reason provided by the contract: 'Auction already ended'. If the transaction failed for not having enough gas, try increasing the gas limit gently."